

Nyx Haile

AI Systems & Research Engineer

xyn@mit.edu | LinkedIn | GitHub | Open to relocation

EDUCATION

Massachusetts Institute of Technology

2026

B.S. Computer Science, B.S. Mathematics, Minor in Economics

Cambridge, MA

- Coursework: algorithms, theory of computation, computer systems security, cryptography, statistics.

EXPERIENCE

Partial

2025 – Present

Cofounder & CEO

London

- Co-founded Partial, a prediction-market exchange where traders take positions on how correlated assets move together.
- Built the Rust pricing engine with my cofounder; it prices two-asset trades over ~200k-cell probability grids in 50-150 ms on a 2-CPU server.

Weiss Laboratory & The SHED, MIT

June 2025 – Sept 2025

Undergraduate Researcher (UROP), generative AI for vascularised liver tissue

Cambridge, MA

- Built the fluidics-design AI agent and wrote the master prompt for a proof-of-concept multi-agent LLM workflow that generates complex vascular STL designs from plain-text specifications; co-inventor on the MIT invention disclosure.
- Wrote a parametric vascular-network generator specification covering stochastic branching, tapering, R-tree collision avoidance and a 0.99 diffusion-coverage loop, sized to a 100x68x38 mm bioprinter envelope.

Vana

Dec 2022 – July 2023

AI Developer

Remote

- Engineered and iterated prompt templates for personalised AI-generated sticker portraits, tuning style and expression cues for identity retention and consistency.

SELECTED PROJECTS

Repositories marked private are available on request.

Strand (private) | Rust, FoundationDB

Capability-authenticated issue tracker

- Built a federable, capability-authenticated issue tracker in Rust on FoundationDB, with line-level blame and content-addressed rationale.
- Shipped a 16-crate, ~71k-line Rust workspace through v1.0.0 GA (10/10 verification gates) and v1.1.0, then a v2.0 governed-execution prerelease.

Forge (private) | Rust, systemd, RPC

Local-first agent control plane

- Built a Rust daemon, operator CLI, Claude Code hooks and provider proxy that coordinate coding-agent work across multiple git repositories.
- Implemented an atomic file-write tool that records attribution for every write and reverts the file if recording fails; the default recorder fails closed.

howtowin.lol | Python, PyTorch, CUDA, Redis

World model for post-game teaching

- Designed a Dreamer-lineage RSSM world model for League of Legends combining static context, player priors and event sequences with outcome, next-event, next-frame and next-decision heads.
- Trained the world model end to end on a 12.8k-game corpus, reaching outcome AUC@15 of 0.88 game-cold and 0.84 player-cold with a clean leak probe.

Sherlock | Rust, Dolt

Token-usage forensics for agent CLIs

- Built a Rust CLI that ingests agent-CLI logs (Claude Code, Codex, Copilot), attributes token usage per session, and reports what burns tokens without adding value.
- Stored ingested usage in a Dolt versioned-SQL repository.

Conch | Rust, ONNX Runtime

Voice-interview-to-brief CLI

- Built a Rust voice-interview CLI that runs spoken Q&A over mic and TTS at three model depths and writes a persisted brief.
- Integrated streaming 16 kHz audio with cloud or offline on-CPU speech-to-text (Nemotron 0.6B int8 via ONNX Runtime) and a live Ratatui waveform view.

TECHNICAL SKILLS

Languages: Rust, Python, TypeScript, SQL **ML:** PyTorch, NumPy, CUDA, GPU training and retrieval **Systems:** Linux/systemd, Docker, FoundationDB, PostgreSQL, SQLite, Redis **Web:** SvelteKit, React, Next.js